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The crustal structure of El Hierro (Canary Islands) from 3-D gravity inversion

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Abstract

El Hierro is the youngest and westernmost island of the Canarian Archipelago. Due to its singular and interesting characteristics, with extensive landsliding, studying its crustal structure can provide very useful information about its origin and evolution. With this aim, we use the gravity inversion methodology obtaining a crustal model of mass distribution. We performed a gravity survey of the entire island, completing the gravity anomaly map of this area with marine data from the United States Geological Survey. The calculated Bouguer gravity anomaly map was studied using a covariance analysis and the least squares technique. To obtain crustal information from these data, we used a 3D gravity inversion based on genetic algorithms (GA). Thus, our inversion technique aims to determine the geometry of the sources of the observed gravity field, upon a prismatic partition of the subsoil volume, and adopting a priori density contrast values. The results of this gravity study of El Hierro Island show the correlation of several volcanic structures with the distribution of the gravity field sources. The characteristic triple rift system of the island is associated with low-density areas, and the older volcanic stages with high-density structures. We also found differences among the structures related to the several landslides, which helped the island takes shape.
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Keywords: inverse problem; gravity anomalies; genetic algorithm; landslides; Canary Islands

1. Introduction

In the Canarian Archipelago (Fig. 1), the island of El Hierro, along with Tenerife and La Palma, appears to have been the most active in terms of volcanic and landslide activity in the recent past (Urgeles et al., 1997). With a surface about 278 km^2 and a maximum altitude of 1501 m in its central region, according to several authors (e.g., Fuster et al., 1993; Carracedo, 1999), this island has been built up from a classic triple armed rift system, which developed according to a geometry of least stress fracture due to upcoming magma. Three well-defined ridges, arranged at approximately 120°, dominate the subaerial morphology of the island, enhanced by large scale landsliding. Many geophysical studies have been performed with onshore and offshore data to better understand this three-armed rift structure and the recent landslide

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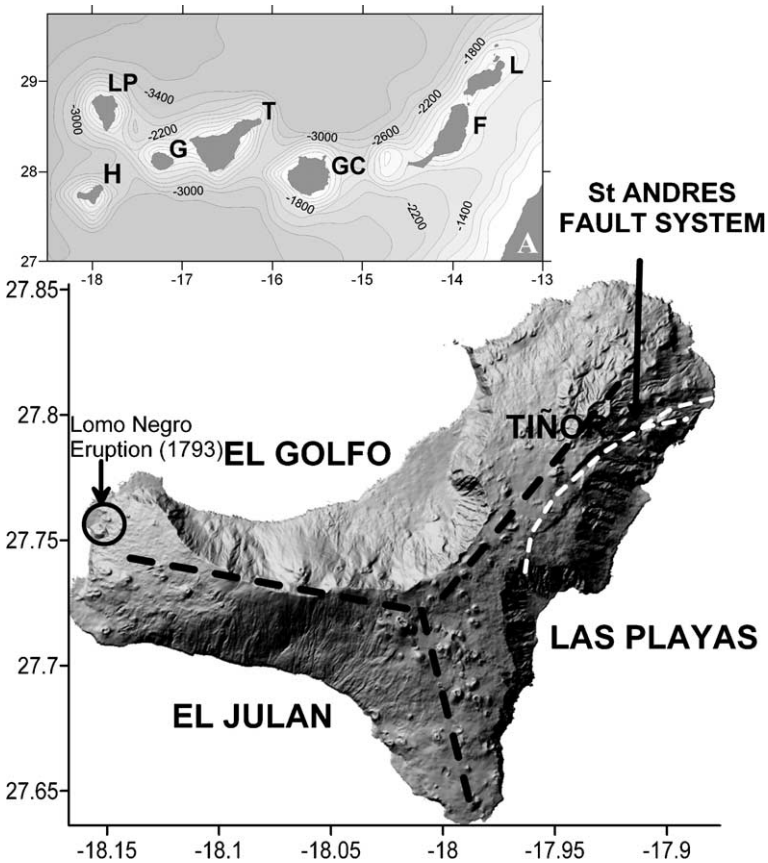


Fig. 1. [Inset] Geographic location of El Hierro island at the northwestern edge of the Canarian archipelago (L: Lanzarote, F: Fuerteventura, GC: Gran Canaria, T: Tenerife, G: La Gomera, LP: La Palma, H: El Hierro; A: Africa). Digital terrain model of El Hierro. Highest altitudes are about 1500 m in the central area. The giant lateral collapses (El Golfo, Las Playas and El Julán) in the island are shown. The dotted line represents the branches of the three-armed rift, where many well-preserved volcanic cones can be observed.

activity in this island. Large-scale landsliding is a key process in the evolution of oceanic islands, and particularly in El Hierro, which is why detailed studies of landslides have been carried out around the Hawaiian Islands (Lipman et al., 1988; Moore et al., 1994), Reunion (Ollier et al., 1998) and the Canarian Archipelago (e.g., Krastel et al., 2001; Watts and Masson, 1995). In El Hierro, several authors have used swath bathymetry and sonar to identify landslide deposits and point out that the active rifts extend into the sea (e.g., Holcomb and Searle, 1991; Masson, 1996; Urgeles et al., 1997; Masson et al., 2002). Onshore studies were carried out to assess the geological and petrological nature of El Hierro (e.g., Hausen, 1972; Pellicer, 1977; Fuster et al., 1993; Carracedo et al., 1997, 2001).

It is firmly established that, of all the geodetic or geophysical techniques available, gravity modelling plays an important role in helping us to understand volcanic structures (Malengreau et al., 1999; Rousset et al., 1989; Loddo et al., 1989; Rymer and Brown, 1986). For example, Kauahikaua et al. (2000) show in their study of Hawaii Islands that the 3D gravity method is valuable for characterizing the basaltic magmatic systems and defining landslide and seismic hazard-related structures. Thus, taking into account previous studies, we conducted a 3D gravity inversion of El Hierro to provide a model of mass density distributions for the crust in this area. We performed a land gravity survey of the entire island incorporating marine gravity data from U.S. Geological Survey (Folger

et al., 1990), in order to better define the gravity map of this area.

The previous studies we performed elsewhere in the Canary Islands led us to a better understanding of their inner structure and evolution. For example, our gravity inversion studies in Gran Canaria (Camacho et al., 2000), Tenerife (e.g., Araña et al., 2000), Lanzarote (Camacho et al., 2001) and Fuerteventura (Montesinos et al., 2005) show the common existence of maximum gravity in areas with the oldest volcanism and minimum gravity near recent volcanic activity. Hence, the experience in this zone provides information to understand the results obtained in this new gravity study with similar volcanism. Moreover, the knowledge acquired through these experiences is used to reduce the well-known limitations of the gravity inversion problem. The most prominent restriction is the non-uniqueness of the solution and, as a consequence, the low deep resolution. These disadvantages can be offset with appropriate tools such as geological information or statistical techniques. In particular, we use the information from similar geological scenarios (elsewhere in the Canary Islands) as a priori constraints on the rock properties and structures expected in the inversion.

The non-uniqueness of the gravity inversion is inherent in the physics of this problem and relates to the existence of a null space, which turns the global minima into a set (of zero misfit to observed data). However, the shape of the search space outside the null space may present local minima that might alter the solution calculated in the inversion problem. When the gravity inversion problem is treated using local optimization methods, e.g. SVD or conjugate gradient, with no a priori information, it can result in a final model heavily biased by the starting model. This problem can be overcome by using a global optimization method, which operates on a randomly selected population of initial models. Thus, we have selected a genetic algorithm (GA) based on evolution and the principles of natural selection. By emulating the processes of selection, mating and mutation, the GA optimizes the model by searching the model space for acceptable solutions and evolving to focus the search on the most promising areas (Holland, 1975). GAs have been successfully applied to many areas within geodesy and geophysics (e.g. Johnson and Wyatt, 1994; Boschetti et al., 1996; Billings et al., 1994; Sen and Stoffa, 1995). However, these algorithms are seldom applied in gravity field studies (e.g Boschetti et al., 1997; Yu, 1995). In contrast to the techniques published by these authors, we have developed an inversion approach based on GA within a 3D context and with the possibility of modeling sources of gravity with diverse physical properties. This methodology has been applied and verified with synthetic examples and smallscale studies, e.g., in archaeology where the crypts identified by means of the inversion methodology were found and excavated, verifying its success (Montesinos et al., 2002). Others studies were achieved in Fuerteventura Island (Montesinos, 2002; Montesinos et al., 2005) where the correlation between the results obtained and other geophysical studies shown the efficacy of this inversion method. This inversion technique was also applied successfully to the island of El Hierro, where it identified structures that help explain the volcanic history of the island. In this paper, we show the results of this last gravity study.

2. Geological setting

The Canary Islands are a volcanic archipelago formed by seven main islands and several islets located at the edge of the western continental margin of Africa, between 27°38′ and 29°25′ latitude N and 13°20′ and 18°9′ longitude W (Fig. 1). The origin of Canaries is still a matter of debate and different authors have put forward various evolutionary models, including the hotspot theory, the coexistence of a hotspot and a constraining complex regional structural framework, etc. (e.g., Wilson, 1973; Schmincke, 1982; Holik et al., 1991; Anguita and Hernán, 1975; Marinoni and Pasquare, 1994; Carracedo et al., 1998). According to the hotspot model, after the initial stage located at the continental–oceanic boundary west of Fuerteventura, the hotspot may have started moving slowly westwards through the Canaries. Presently, it is presumably located under the island of El Hierro (Holik and Rabinowitz, 1992; Carracedo et al., 1998, 2001). El Hierro and La Palma (Fig. 1) are the youngest and westernmost islands of the archipelago (Abdel-Monem et al., 1971) and, along with Tenerife, seem to have been most geologically active in the recent past (Urgeles et al., 1997). At present,

they are at the shield stage of growth and the rest of the Canary Islands are already at the posterosional stage (Carracedo, 1999). The most recent massive landslide probably took place in El Hierro (El Golfo) 15 Kyr ago. Additionally, this is the only island of the archipelago that has an aborted and incomplete lateral collapse on its eastern flank (Day et al., 1997).

This special evolution of El Hierro, marked by the occurrence of major landslides, may be related to the existence of active Y-shaped rifts, with the NE and NW branches more active than the S branch. This island was probably formed over a stationary source of magma presenting a perfect concentric development, with superimposed volcanoes and a regular three-armed rift geometry (Carracedo, 1996). However, new offshore data do not support a simple interpretation for the volcanic rift arms. Instead of discrete ridges of spaced volcanic rift arms, Gee et al. (2001) observed (to the NW and NE) broad areas of irregular morphology, suggesting that rifting activity might not be confined to narrow zones. Furthermore, they suggested that the anomalously long and steep-flanked Southern Ridge could be part of an older, eroded volcanic edifice that predates much of the other submarine flanks of El Hierro. Thus, the Southern Ridge has a distinctive gullied morphology, which strongly contrasts with adjacent flanks. Furthermore, the Southern Ridge did not form by continuous dyke intrusion from the El Hierro volcanic centre, but is a stable edifice, relative to the rest of El Hierro.

The rift zones, combined with the instability in the volcanic islands, progressively tend to increase the possibility of giant landslides that appear as steep cliffs and form arched bays (Cantagrel et al., 1999). These landslide processes on volcanic islands have been researched extensively (e.g., Masson et al., 2002; Ollier et al., 1998; Urgeles et al., 1997; Moore et al., 1994).

Three giant lateral collapses can be seen in the onshore geology of El Hierro (Carracedo et al., 1999): the Tiñor, El Julán and El Golfo (Fig. 1). The San Andrés fault on the NE flank of the island may correspond to an aborted lateral collapse (Day et al., 1997) and the embayment of Las Playas, on the SE flank of the island, may be a similar feature (Fig. 1).

The eruptive history of El Hierro, which is still in its juvenile stage of shield growth, is probably the best constrained geochronologically of any Canary Island (Carracedo, 1999; Fuster et al., 1993). The three main volcanic units associated with successive volcanic edifices and directly related with the giant landslides are: (a) Tiñor volcano, (b) the El Golfo edifice, (c) the rift volcanism. Following the geochronology established by Fuster et al. (1993), three periods of activity may be distinguished. The so-called "A-Ancient series" outcrops in the NE of the island and in a high area of El Golfo. The "B-Ancient series" appears on the surface in the walls of El Golfo, Las Playas and on the western coast of the island. These series predominate in the central part of the island, where three structural axes intersect, and correspond to the shield stage, according to the volcano-stratigraphic units defined in Hawaii (Walker, 1990; Carracedo et al., 2001). Fuster et al. (1993) postulated the existence of a first emission period from independent edifices that remained isolated. In the beginning, they may have been joined, but were separated by destructive phenomena, producing the emission of B-Series. The eruptive centres were dispersed, forming a bigger shield volcanic edifice than the original structure. The main emission centre of this edifice may be near the centre of El Golfo depression. This formation included giant landslides, which created the great depressions of El Golfo and Las Playas.

The intermediate and recent series (posterosional or rejuvenated stage) were established to distinguish the eruptive materials overlapping the ancient series. Some volcanic edifices of the recent series are well preserved, such as Soliman volcano (near of San Andrés). The only historic eruption could have occurred in Lomo Negro, on the western coast of the island (Hernández-Pacheco, 1982) in 1793.

Carracedo et al. (1999) and Carracedo et al. (2001) give a detailed description of the geology and volcanology of the island of El Hierro, revising the results established by Fuster et al. (1993). The first subaerial volcanic activity in El Hierro formed a steep volcano that developed between 1.12 Ma and 882 Ka (Gillou et al., 1996) in the eastern part of the island. This volcano was affected by a northwest-directed giant collapse; the scar thus formed has been largely buried by later volcanism. The Tiñor collapse probably fol-

lowed the late explosive episodes of the volcano that occurred about 882 Ka. Subsequently, another volcanic edifice, the El Golfo volcano, completely filled the Tiñor collapse embayment. When this volcano was well developed, the El Julán collapse, which destroyed the SW flank of the El Golfo volcano, probably occurred. El Julán is a 16,000 m wide embayment, completely covered by lava from vents located at the summit of the NW rift. Holcomb and Searle (1991) identified this feature as a gravitational collapse from offshore GLORIA sidescan sonar images.

On the SE flank of the island, the Las Playas landslide is interpreted as a debris avalanche that appears to result from the partial failure of the upper flank (Gee et al., 2001).

The El Golfo embayment is perhaps the most spectacular feature of all on El Hierro. It is some 15,000 m wide, extends some 10,000 m inland, and its headwall is more than 1400 m high in places. Conclusive offshore evidence identifies the El Golfo embayment as the most recent gravitational collapse in the Canaries (Masson, 1996; Canals et al., 1997; Urgeles et al., 1997; Masson et al., 2002; Bravo, 1982) although its age is still a matter of debate. One possible means of reconciling the contradictory onshore and offshore evidence has been postulated by Carracedo et al. (1999), who argue that there were two lateral collapses: one largely subaerial and occurring about 130 Ka ago, and the other affecting the seaward parts of the lava platform built up within the embayment, and also the submarine slope of the island down to considerable depths.

The rift volcanism forms a relatively thin sequence with emission vents distributed in three volcanic rift branches that have not as yet produced a topographically distinct edifice. Rifts are defined by the concentration of eruptive vents.

With respect to the thickness of the crust, several authors have used seismic measurements and free-air anomalies to study this feature in El Hierro. Bosshard and MacFarlane (1970) found that the depth of the Mohorovicic discontinuity (Moho) is about 14,000–15,000 m beneath El Hierro. Watts (1994) calculated a value of 3000–4000 m for the lithospheric flexure caused by volcanic loading beneath this island. Taking into account this value, he estimated a Moho depth of around 15,000-16,000 m.

3. Topographic and gravity data. Analysis of gravity anomalies

To calculate the gravity data terrain correction, we use a 1 : 25,000 scale digital terrain model provided by the National Geographic Institute of Spain. Fig. 1 is an orthographic view of the terrain model where the highest altitudes are around 1500 m in the central area of the island. The figure clearly displays the great depressions of El Golfo, Las Playas and El Julán. The topographic information is supplemented with ETOPO5U (National Geophysical Data Center, 1988) bathymetric data.

We consider two gravity data sets from different sources. The first source is the observations we conducted in 2003, with a LaCoste and Romberg gravity meter with digital electronic readouts. We measured a total of 93 stations on El Hierro, with a nearly homogeneous distribution and located at average intervals of 1600 m (Fig. 2). The gravity station coordinates were obtained by differential GPS. These gravity data were corrected with the tidal model calculated for the Canarian Archipelago using tidal gravity observations made in the islands of Tenerife and Lanzarote (Arnoso et al., 2000, 2001). Adjusting the data with 40% repeat observations and correcting for drift, jumps and tides, produced residuals with a standard devia-

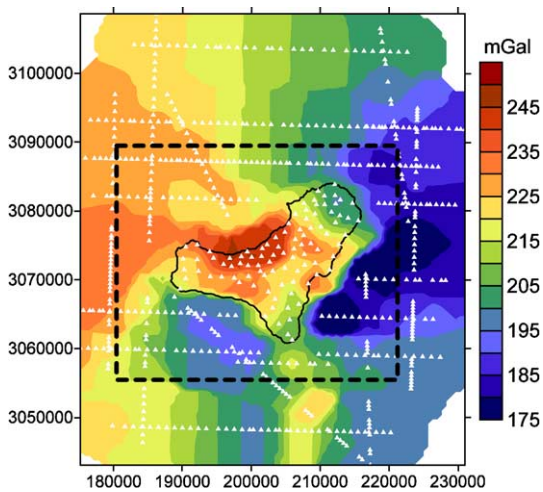


Fig. 2. Distribution of gravity stations and Bouguer gravity anomaly map. The value used for the density of terrain correction is 2510 kg m^{-3}. Contour interval is 5 mGal. The dotted rectangle shows the zone used in the inversion process. Maps are referenced to UTM coordinates.

tion of 0.030 mGal (1 mGal=10^{-5} ms^{-2}). The second source of data used to detect the regional trend in the gravity, was marine gravity data provided by U.S. Geological Survey (USGS) collected from a cruise in 1987 aboard the Starella Research Vessel (Folger et al., 1990), up to 10,000–25,000 m offshore of the island (Fig. 2).

Gravity values from both data sets are referenced to International Gravity Standarization Net, IGSN 1971. To calculate the Bouguer anomaly map, we use the 1980-normal gravity formula, and determine the density value to apply the terrain correction, extended to 25,000 m away from each station, according to the regularity of the effect beyond this distance. To select a suitable average terrain density, we choose a value that reduces gravity anomaly field distortion. This value produces a minimum of the root mean square (rms) error, using a covariance analysis of the Bouguer anomaly data and further least-squares prediction (or signal filtering) (Moritz, 1989; Montesinos, 2002). This process is applied to several density values to determine which is the best. The gravity anomaly map is calculated for each tested density value. By computing the covariance function for these data, we predict the gravity signal at observation points with the least-squares method (Moritz, 1989). The difference between these values and the observed gravity (residual) is used to detect erroneous observations. If the residual in one station is more than twice the standard deviation, this observation may be erroneous and is rejected and the process is repeated, applying the same removal criteria according to the new standard deviation. With the final residuals thus obtained, we calculate the rms of these data, checking the quality of data for the density value used for the terrain correction. Then, using this methodology to look for the minimum rms, we select a mean value of 2510 kg/m^3 for the density of the terrain masses and a value of 1027 kg/m^3 for the seawater. This terrain density is as near to the value of 2470 kg/m^3 as can be obtained by Nettleton's method. Moreover, this value is in line with the density calculated for the other islands with similar volcanism: 2500 kg/m^3 for Fuerteventura (Montesinos et al., 2005), 2480 kg/m^3 for Lanzarote (Camacho et al., 2001), 2300 kg/m^3 for Gran Canaria (Camacho et al., 2000) and 2320 kg/m^3 for Tenerife (Araña et al., 2000). Taking into account that this value is an average density for several regions of the island, the results in the inversion process may be influenced by an excess- or deficit-of terrain correction. This effect occurs in the areas where the terrain density is, respectively, less or greater than the density considered and will be reflected in the local gravity anomalies field.

The aforementioned covariance analysis also serves to establish the consistency and homogeneity of the different data sets. If the information provided by the two data groups is not uniform throughout in composition or structure, the data are not homogeneous and a suitable covariance function cannot be determined. Furthermore, from this analysis a covariance matrix related to the inaccuracies for the gravity data can be calculated, which is useful in the following inversion process (e.g., Camacho et al., 2000).

Covariance analysis of the gravity data and a further least-squares prediction permits the relation between the observed data and the estimated signal to be given in terms of the covariance and that the noise and the signal to be uncorrelated. By using a correlation analysis of the data and assuming that the covariance between two stations depends only on the horizontal distance, we calculate a distribution of empirical covariances, which is adjusted with a suitable analytical covariance function (Barzaghi and Sanso, 1983). The usual formulae for least-squares prediction (Moritz, 1989) give us a first estimation of the signal at this level of covariance and the corresponding prediction error matrix. By considering the residual values (observed data minus estimated signal), which is the non-correlatable part for the previous covariance level, we repeat this process with a new covariance analysis in order to detect, if possible, correlated signals of minor amplitude. This procedure ends when no further signal can be detected. Finally, by removing these detected signals from observed data, the distribution of the resulting residual values displays the non-correlatable noise. In this case, we estimate the gravity signal of three successive levels defined by their covariance functions (Fig. 3). The sum of the three components of the gravity signal is the correlated part of the data, and its residual relative to the observed data is the non-correlated noise, which shows a standard deviation of 0.577 mGal (Fig. 4). This may be observational noise, very local anomalies or uncertainties in the previous corrections (such as terrain corrections). Besides, this value attests to the

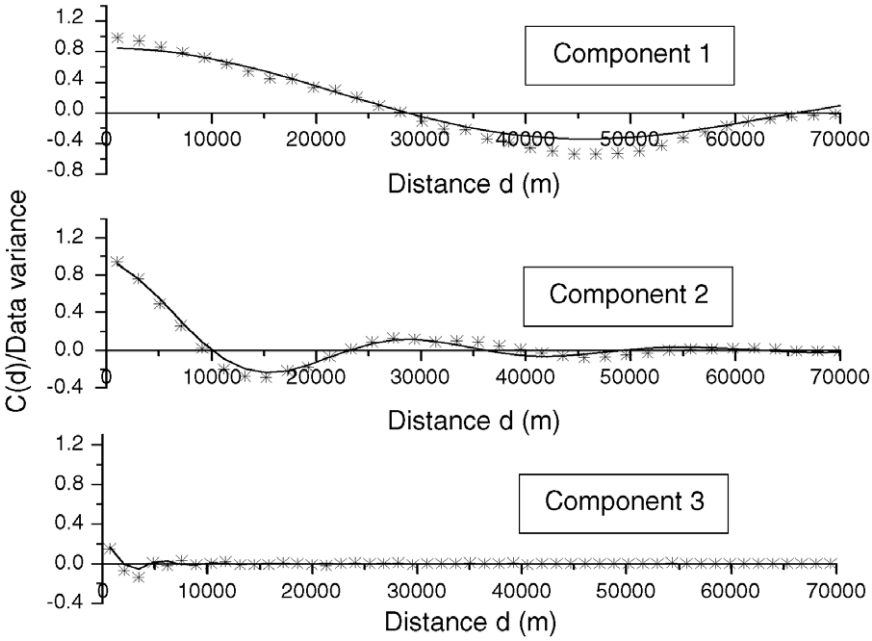


Fig. 3. Covariance functions adjusted (line) to the three successive empirical covariance distributions (stars).

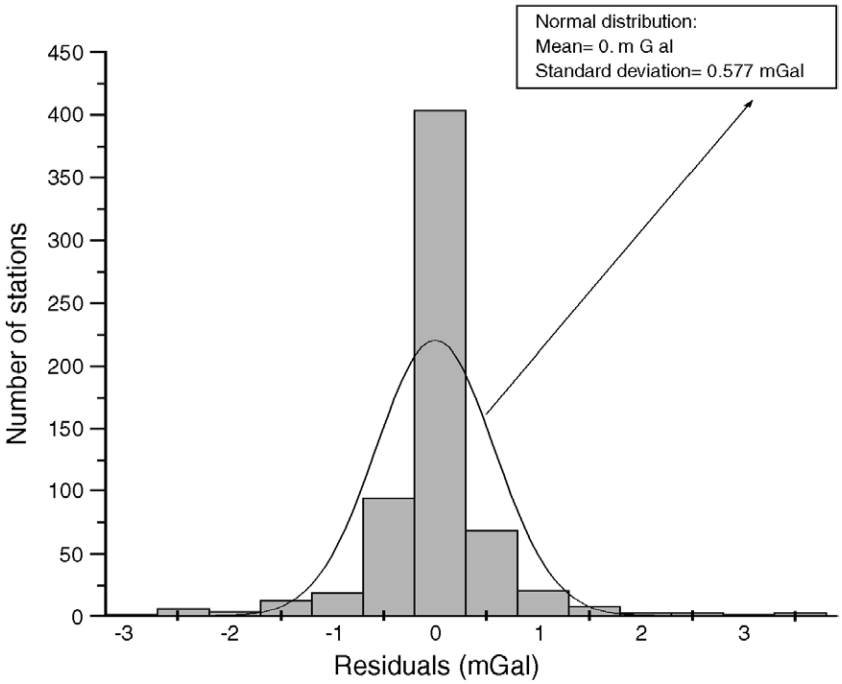


Fig. 4. Histogram of the residuals of the filtering process. This non-correlated noise of the data follows a normal distribution of mean 0. mGal and standard deviation 0.577 mGal.

good quality of the data, as required for a study of this type, and to the successful homogeneity seen between the data sets in the determination of the covariance functions. The error matrix of the estimated signal calculated from the covariance analysis will be considered an error matrix for the data, E_{ss}, which is used in the inversion process that follows.

Using the described criteria to remove the outliers and selecting marine stations a minimum of 1000 m apart, ultimately we select 257 points from the USGS data bank together with the entire set of 93 stations in El Hierro to obtain the Bouguer anomaly map of the region (Fig. 2). The gravity anomaly map ranges by 80 mGal, with maximum values in the El Golfo area and minimum values in the southeastern offshore area. To interpret this map, we take into account the data gaps that exist mainly offshore and affect the gravity anomalies, which appear extrapolated in these areas.

The Bouguer gravity anomalies map (Fig. 2) is dominated by maximum values onshore except on the western and southern edges of the island. Although the offshore prolongation of the main maximum seems to be located towards the north-western area, this is not well defined because of the data gap near the coast. The marine gravity data indicate the eastern area dominated by a strong gravity minimum with lowest values in the area occupied by the Las Playas landslide. These data also give minimum values in the area of El Julán landslide (see Fig. 1). However, the offshore continuation of the NW and S ridges coincides with maximum gravity values.

4. Gravity inversion method

There are two general approaches for the inversion of gravity anomalies: linear and non linear. In the nonlinear case, a known density contrast is assumed and adequate algorithms are designed to determine the geometry of the anomaly's source. In the linear case, the density of the source with known geometry is determined. Many techniques have been developed to solve both problems. However, the inverse modeling schemes based on genetic algorithms (GA) have gained popularity over the last decade. GA consist of search methods modeled on the evolutionary behavior of biological systems, according to Darwin's evolution principle. They can solve complex, non-linear optimization problems without the need for starting models or curvature information, simply by direct sampling of the solution space. A classical GA was proposed by Holland (1975) and described in Goldberg (1989). Unlike other techniques, GA work with a population of models (individuals) and selecting an initial set of models at random, they seek to improve the fitness to the observed data generation after generation, in an evolutionary process.

Following the GA principles, we put forward the gravity inversion problem as an evolutionary process where the individuals are the possible models that represent the sources of the observed gravity anomaly field (anomalous bodies). We have already successfully applied this model in other studies (Montesinos, 2002; Montesinos et al., 2002, 2005), where a detailed explanation about its formulation can be found. This inversion approach seeks the geometry of the sources of the gravity field, adopting the hypothesis that these structures are characterized by θ prescribed mass density contrasts, $d=\{d_1, \ldots, d_q\}$. The models are defined from a fixed discrete 3-D partition of m prismatic elements of the subsurface volume close to the survey. Thus, the anomalous sources are constructed by assigning to each prism one of the θ prescribed density contrasts. In this case, our GA works over a set of models (a population of individuals), making successive modifications in an iterative process, seeking a model that minimizes the discrepancy between the gravity field generated by the model and the observed gravity data (error function). In the t-step of the iterative process, each individual is characterized by a vector (or chromosome)

$$m_k^t=\left(\rho_1^{kt}, \rho_2^{kt}, \ldots, \rho_m^{kt}\right)^T \text{ with } \rho_j^{kt}\in\left\{d_1,\ldots,d_q\right\} 1\leq j\leq m;$$

where T indicates transpose, and

m_k^t k-model or potential solution of the problem in the t-step

ρ_j^{kt} Density assigned to the j-prism of the k-model in the t-step

$\{d_1,\ldots,d_q\}$: Set of density contrasts allowed in the model.

To simplify the notation, we use m_k and ρ_j^k instead of m_k^t and ρ_j^{kt}, assuming that we are in the t-step of the process.

The model parameters are the densities of each prism of each individual of the population. They must be selected from a discrete set of values established a priori from geological or geophysical information. We look for the geometrical properties of anomalous bodies from the distribution of the density contrasts assigned to the prisms, in a nonlinear context for the inversion problem.

In the inversion method, density values may be invariant during the process (a data set, with several positive and/or negative values) or may change throughout the procedure, more feasible density contrasts being added as the structures become better defined (e.g., Cordell, 1973; Montesinos et al., 2005). However, we must always consider a finite number of values.

Let us consider n gravity stations P_i (x_i, y_i, z_i), $i=1, \ldots, n$, not necessarily gridded, located in a rugged topography and with observed gravity anomaly values g_{obs}. The gravity attraction, A_{ij}, at the i-th station P_i (x_i, y_i, z_i), due to the j-th prism, for unity density, is given by (Pick et al., 1973):

$$A_{ij} = -G\left[\left[\left[x\log\left(y+\left(x^2+y^2+z^2\right)^{1/2}\right)\right.\right.\right.$$
$$+ y\log\left(x+\left(x^2+y^2+z^2\right)^{1/2}\right) + z\ atan$$
$$\left.\left.\left.\times\left(z\left(x^2+y^2+z^2\right)^{1/2}x^{-1}y^{-1}\right)\right]_{u_1^j-x_i}^{u_2^j-x_i}\right]_{v_1^j-y_i}^{v_2^j-y_i}\right]_{w_1^j-z_i}^{w_2^j-z_i},$$

where G is the gravitation constant, the edges of the j-th prism are parallel to the reference axes, and the limiting coordinates for its volume are: u_1^j, u_2^j for the x coordinate, v_1^j, v_2^j for the y coordinate and w_1^j, w_2^j for z coordinate. The matrix $\mathbf{A}$, with components A_{ij}, is the design matrix and contains the effect of the rugged terrain, the stations distribution, subsurface partition, etc.

According to the direct problem, the gravity anomaly in the i-th station P_i, $i=1, \ldots n,$ produced by the model m_k is

$$g_i^k = \sum_{j=1}^{m} A_{ij}\rho_j^k. \tag{1}$$

We define the error function which must be minimized and determines the adjustment of each individual by (Montesinos et al., 2005)

$$F(m_k) = (\mathbf{A}m_k - g_{\text{obs}} - G_m\boldsymbol{u})^T E_{\text{ss}}^{-1}(\mathbf{A}m_k - g_{\text{obs}} - G_m\boldsymbol{u}) + \beta m_k^T \mathbf{C}_{\text{M}} m_k, \tag{2}$$

where $\boldsymbol{u}$ is a unitary vector, β is an empirically determined parameter corresponding to the Tikhonov regularization technique (Schwarz, 1979), and the $m \times m$ matrix $\mathbf{C}_{\text{M}}$ describes the uncertainty of the model. A diagonal matrix can design this, with its elements defined by the inverse of the diagonal elements of $\mathbf{A}^{\text{T}}$ E_{ss} $\mathbf{A}$. So, we assign a value relative to the gravity attraction that it produces at the stations to each prism, defining in this way its uncertainty. The data error matrix E_{ss} can be obtained from previous covariance analysis (Moritz, 1989).

The process starts with a population of np models, $P(0)$, which can be generated randomly or, to avoid premature convergence to a non-suitable model, it can be started with all models identical, without information, where all the prisms have null density contrasts (empty models),

$$P(0) = \left\{m_1^0, m_2^0, \ldots, m_{np}^0\right\} \qquad m_k^0 = \left(\overbrace{0, 0, \ldots, 0}^{m}\right)^T$$
$$k = 1, \ldots, np.$$

The independence between the solution and initial population corrects these possibilities in GAs. If we select a null population, high variability within the population is required in the first stages of the evolution process. However, the inheritance of the best individuals must be preserved. These conditions are achieved using mutation and crossover operators. Schaffer and Eshelman (1991) compare these operators and affirm that a good solution cannot be reached without the crossover.

The size of the population, np, is selected as a function of the quantity of prisms, data, etc., (generally a value of 10 gives successful results, although another selection would not alter the results but instead the convergence rate) and is invariant throughout the process. In the iterative procedure, the individuals of the population are evaluated (error function Eq. (2)) and can be chosen (selection operator) to continue the evolutive process. Then, two genetic operators (mutation and cross) act on these selected

models producing ℓ new individuals, which are also evaluated. Then, the selection operator chooses, from these np (old)+ℓ (new) models, the best np individuals, and the iterative procedure is repeated.

The selection operator (Michalewicz, 1994) acts according to a cumulative probability q_k, for each model m_k, defined by

$$q_k = \sum_{i=1}^{k} \frac{F_{\mathrm{T}} - F(m_i)}{\sum_{j=1}^{np} \left(F_{\mathrm{T}} - F(m_j)\right)}$$

where F_{T} is the adjustment of the whole population, $F_{\mathrm{T}} = \sum_{k=1}^{np} F(m_k)$. To decide which models will continue the evolution process, this operator generates np times a random number $r \in [0,1]$, if $q_{k-1} < r \leq q_k$ for some $k=1, \ldots, np$ ($q_0=0$), then m_k is selected. Obviously, the best models obtain most copies.

To create new models, the mutation and crossover operators act on these selected individuals. The mutation provides random diversity in the population according to a predefined probability p_{m}. This operator assigns to each m_k, $k=1,\ldots,np$, a random number $r \in [0,1]$. If $r > p_{\mathrm{m}}$; this model does not change any of their prisms. Otherwise, if $r \leq p_{\mathrm{m}}$, some prisms will mutate, randomly changing their density contrast value among the θ fixed density contrasts. If the model m_k must mutate, for each j-prism ($j=1, \ldots, m$) the accumulative error, e_j^a, is defined by

$$e_j^a = \sum_{i=1}^{j} \frac{e_{\mathrm{i}}}{\sum_{l=1}^{m} e_l}, \quad \text{and} \quad e_0^a = 0.$$

Then, a random number $s \in [0,1]$ is generated: if $e_{j-1}^a < s \leq e_j^a$ the density of j-prism changes. Traditionally, only one element (prism) is mutated (Grefenstette, 1986) but in this case, there may be a very large number of elements, so we repeat this process twice for each individual selected for mutation. Although most authors (e.g., Reeves, 1993; Grefenstette, 1986; Michalewicz, 1994) choose very low mutation rate values (about 0.01), because they usually work with few unknowns, we suggest a value of $p_{\mathrm{m}}=0.3$ (this value can be increased if the amount of unknowns increases).

The next operator exchanges genes between two individuals with probability p_c (e.g. Michalewicz, 1994). For each individual, m_i, we generate a random number $r_i \in [0,1]$. If $r_i < p_{\mathrm{c}}$, then m_i is selected to cross. Coupling selected models (m_i, m_k), where

$$m_i = \left(\rho_1^i, \rho_2^i, \ldots, \rho_\ell^i, \ldots, \rho_m^i\right)^{\mathrm{T}} \quad \text{and}$$
$$m_k = \left(\rho_1^k, \rho_2^k, \ldots, \rho_\ell^k, \ldots, \rho_m^k\right)^{\mathrm{T}},$$

the operator exchanges the genes from the position $\ell \in [1,m]$ selected at random:

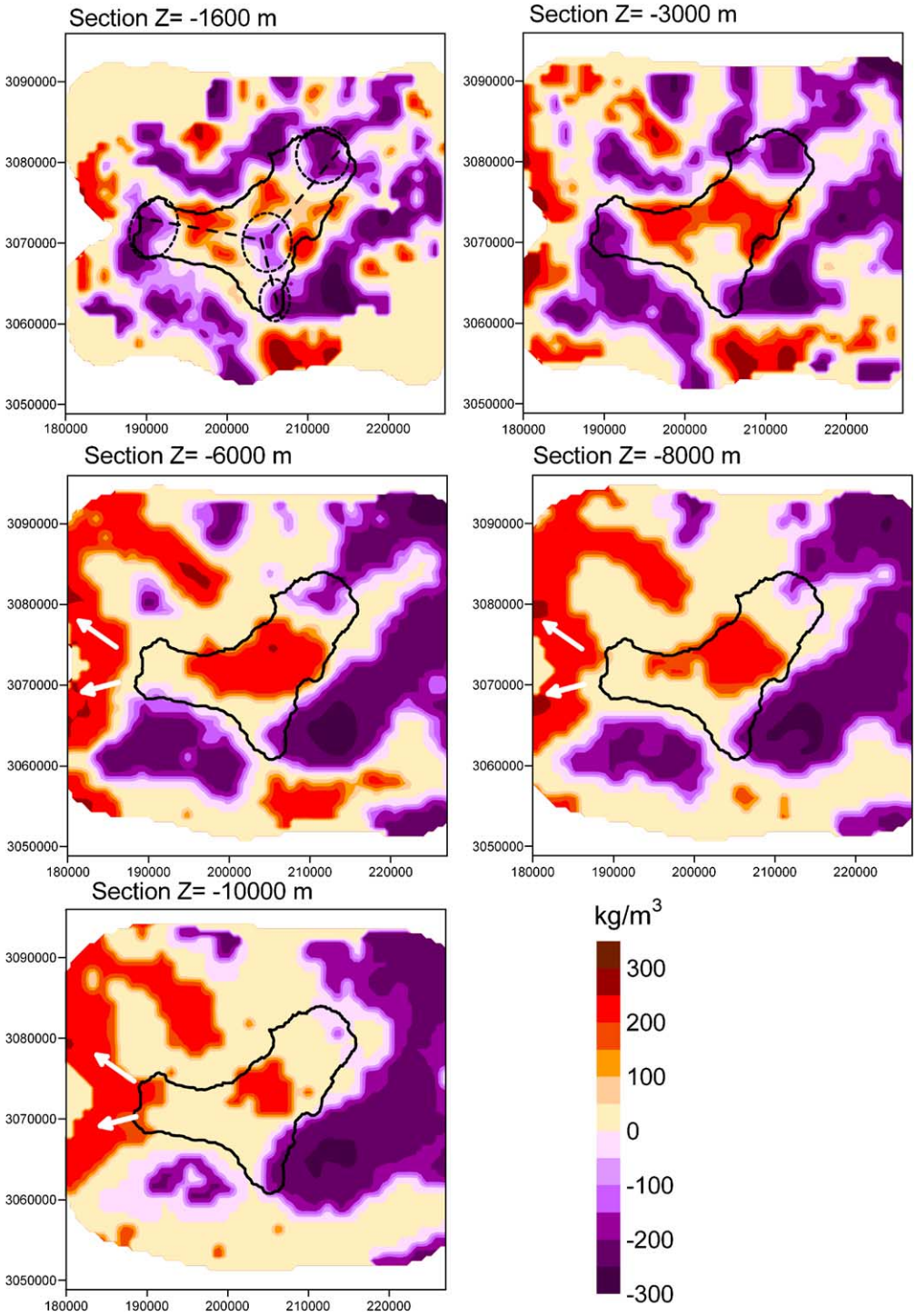
$$m_i = \left(\rho_1^l, \rho_2^l, \ldots, \rho_\lambda^\kappa, \ldots, \rho_\mu^\kappa\right)^{\mathrm{T}} \quad \text{and}$$
$$m_k = \left(\rho_1^k, \rho_2^k, \ldots, \rho_\ell^i, \ldots, \rho_m^i\right)^{\mathrm{T}}.$$

We use a crossover rate value, p_{c}, of about 0.6, as is recommended in GA literature (e.g., Grefenstette, 1986; Michalewicz, 1994).

The modified models are evaluated again with the unmodified ones, repeating the evolutionary process (selection/mutation/crossover) until a model that minimizes the error function is found. After numerous generations, GA convergence slows greatly. In order to establish to an end this stage of the process, we introduce a predetermined value according to the accuracy of the data. The following stage consists of obtaining a plausible model, as a solution to the inverse problem, which agrees with the geological reality.

The solution to the inverse problem poses some uncertainty because of the non-uniqueness of the gravity inversion and the possibility of deviation to local minima in the search space. For example, the adjusted geometry for the sources of the gravity field bodies will depend on suitable selection of the density contrasts (positive and negative) which implies some ambiguity. To produce a model with a smoother geo-

Fig. 5. Adjusted model of density contrasts obtained by means of 3-D gravity inversion. The density contrast values used in the model correspond to a discrete set of values between −300 and 300 kg m^{-3}. Several horizontal sections, from 1600 to 10,000 m depth, are shown. At greater depths the model does not shows any structure. In the shallowest section, low-density bodies are predominant over the triple arm rift (dotted lines) that characterizes this island. These onshore structures with negative density contrasts are marked with dotted circles in this section. In the deeper sections, white arrows indicate the bifurcation of the onshore NW rift zone seaward according to Gee et al. (2001).


Section Z= -1600 m
Section Z= -3000 m
Section Z= -6000 m
Section Z= -8000 m
Section Z= -10000 m
3090000
3080000
3070000
3060000
3050000
180000
190000
200000
210000
220000
kg/m3
300
200
100
0
-100
-200
-300

metry so as to remove this uncertainty and avoid local minima, we employ a smoothing technique equivalent to minimizing the whole anomalous mass (Montesinos et al., 2005). This operator acts in each model, when the error function is minimized, assigning each prism an averaged value of the density contrasts of the adjacent prisms. Then, these smoothed new individuals are evaluated again. If the best model fits the observed gravity data successfully according to the predefined limit, or does not improve the previous models after several stages, the inversion ends; otherwise, the evolutive process continues.

To apply this methodology to the island of El Hierro, taking into account that contiguous stations are some 1600 m apart and the mean diameter of the survey is about 35,000 m (and the computer limitations), we partitioned the local subsurface into 14,768 rectangular prisms, with sides ranging from 1000 m for the shallow elements (1000 m depth) to 1200 m for the deepest blocks (at 14,000 m depth). According to the suggested method, we selected for the whole subsurface volume, the possible density contrasts by means of the extreme values, -300 kg m^{-3} and $+300$ kg m^{-3}, and an exponential variation law of contrasts. These values have been selected empirically to obtain connected and well-developed volumes, producing an inversion model with a negative mean density contrast of 210 kg m^{-3} and a positive one of 250 kg m^{-3}, which could correspond to the usual contrasts among structural elements of the Canarian volcanism. Bosshard and MacFarlane (1970) deduced density values of 2300 kg m^{-3} for sedimentary material, 2600–2700 kg m^{-3} for crustal material, and 3100 kg m^{-3} for mantle-type material. From these values, the selected density contrasts seem to be credible for this volcanic island. When different density contrasts are used, the resulting solutions are rather similar, although for higher contrasts, smaller volumes would be obtained. In the gravity studies conducted elsewhere in the islands, we selected similar density values: (-300, 200) kg m^{-3} for Tenerife (Araña et al., 2000), (-340, 340) kg m^{-3} for Lanzarote (Camacho et al., 2001), and (-300, 400) kg m^{-3} for Fuerteventura (Montesinos et al., 2005). This result is coherent with the similar volcanism that took place in these islands, and with the fact that Tenerife and El Hierro are, at present, at the same stage of development (Carracedo, 1999).

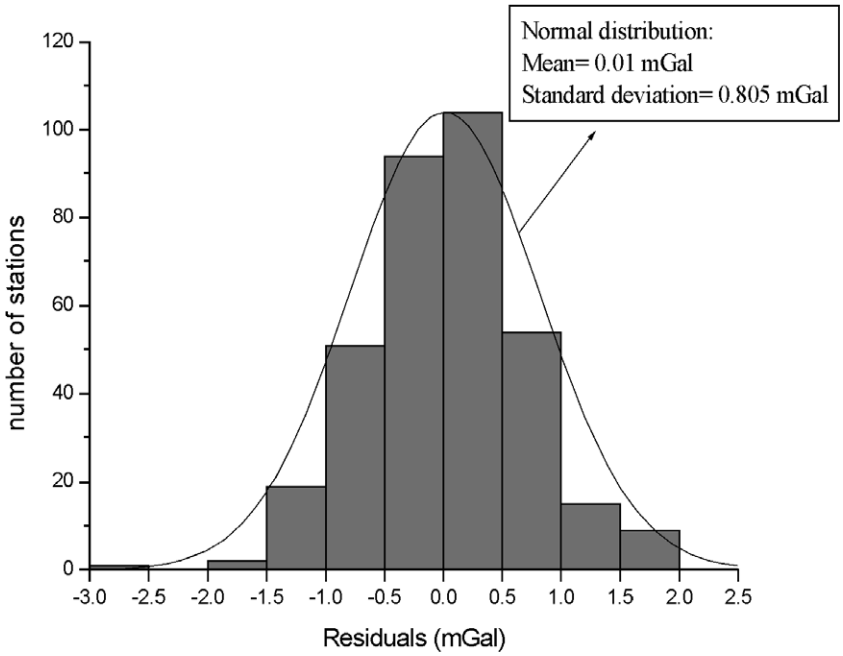


Fig. 6. Distribution of residuals corresponding to the inversion process, with a standard deviation of 0.805 mGal, close to the value obtained for the data noise.

The adjusted local model of density contrasts is shown in several horizontal sections, to several depths with respect to sea level (Fig. 5). Some interesting features can be highlighted from this model with a view to the correlation with the geological and volcanic information. Finally, Fig. 6 shows the distribution of residuals of the inversion adjustment, which have a standard deviation of 0.805 mGal, in accordance with the data accuracy. In this case, we stopped the evolution process when the fit of the best model at that stage failed to improve after applying two successive smooth operators.

5. Results and discussion

The gravity field sources model obtained by means of inversion shows a predominant distribution of positive density contrasts for the onshore structure of El Hierro. This distribution of high-density bodies occupies broad areas with fairly irregular morphology and, a priori, they do not follow a scheme that suggests that the rifting activity detected in the island might be confined to narrow zones. This fact seems to agree with the idea formulated by other authors, who suggest that oceanic islands develop along rift zones that feed from central vents. Macdonald (1949) was one of the earliest authors to indicate these circumstances in Hawaii, and Gee et al. (2001) established for El Hierro that the growth of the submarine part of the edifice was not focused within discrete narrow zones, but over a much broader area. Thus, the only high density body to appear in the deepest sections of the model (horizontal profiles, 6000–10,000 m depth) may be related to the ancient extrusion of massive basaltic material that forms the main structure of the island. This material lies near the zones where the oldest volcanism outcrops (Fuster et al., 1993) and could be the feeding system of the Tiñor and the El Golfo volcanoes. This agrees with the fact that the present outcrops of the first volcano are confined to the NE flank of the island and inside the Las Playas embayment, and the El Golfo volcano subsequently completely filled the Tiñor collapse embayment (Carracedo et al., 1999). This high-density body occupies a nearly central position slightly displaced towards the NE and may be a part of the shield volcanic edifice. A similar structure has been identified for the other islands of the archipelago where gravity inversion techniques have been applied (Araña et al., 2000; Camacho et al., 2000, 2001; Montesinos et al., 2005). It is displaced slightly towards the North, indicating that the main activity in the early stages of development of the island was concentrated in the north, and therefore related to the NW and NE rifts.

The shallowest sections of the model must be cautiously reasoned because of the distance between contiguous stations (nearly 1600 m). This fact determines the minimum depth of inversion model (e.g., Fedi and Rapolla, 1999). However, it must be noted that the stations are considered in their original position on the topographic surface, which is 1500 m high in the central zone. Thus, we can consider sections from nearly 1600 m. In these shallowest sections of density contrasts, alignments of low-density bodies can be identified near the triple armed rift referenced by several authors. They are located in the central area and on the edges of the arms (W, NE and South of the island) (Fig. 5). These low-density crustal structures are justified by the concentration of recent (Quaternary) eruptive vents and the relatively recent associated fractures. These central minima are located over the main high-density structure found in the deep model, which is associated with the old Tiñor edifice and the feeding system of the early volcanic activity of the zone.

The minimum density alignments on the W, NE and South edges of the island are located in the same place as the Recent Series activity (e.g., Fuster et al., 1993). The western coast of the island, where the single historic eruption from the Lomo Negro vent took place in 1793, is characterized by negative density contrast relative to surrounding areas (Figs. 1 and 5).

There are several high-density bodies at depths of between 1600 and 3000 m that appear as independent structures at shallower sections, and join as they go deeper. These bodies are isolated in the area where there is an abundance of intruded basaltic dykes exist (Carracedo et al., 2001). However, it could be also conditioned by the rift system and the recent volcanism that took place over the NE and NW volcanic rift arms. Moreover, on the SE flank of the island, the border of the positive density contrast structures matches the San Andrés fault system (Figs. 1 and 5).

The structures located in the offshore areas of the model must be interpreted with caution because they

are not supported by much data, and a clear border effect exists. However, if there are high- or low-density structures, they will be identified, at least in general lines, by the inversion process.

In the south of the model, another high-density body is located between the surface and a depth of 6000 m. This structure is related clearly to the eroded volcanic edifice associated by Gee et al. (2001) with the Southern ridge. In accordance with the hypothesis of these authors, this body is independent of the rest of the island.

It is remarkable that the areas occupied by El Julán and Las Playas landslides correspond to low-density material. However, the area where the El Golfo landslide occurred is mainly composed of high-density structures. This contrast underscores the distinction between the landslides of El Golfo and El Julán and Las Playas. The differences are conditioned by the nature of the landslide material (e.g., volcanic or intrusive rocks, pyroclastic deposits or volcaniclastic sediments) and the flow processes (e.g., flow or avalanche, speed of emplacement) and, in this case, are fully documented by offshore data (e.g., Masson et al., 2002). The El Golfo landslide is categorized as a debris avalanche. It has a typically blocky morphology, with individual blocks up to 1 km or more in diameter scattered randomly across the avalanche surface. This avalanche affected only the superficial part of the island volcanic sequence (Masson et al., 2002). Thus, at the greater depths of the model, consolidated bodies with high density appear associated with the oldest volcanism and, at shallower sections, a variety of low-density materials appears to be combined with other denser small structures. Las Playas and, possibly, El Julán landslides might tentatively be identified as slumps, affecting a considerable thickness of volcanic edifice (up to 10,000 m thick on the Hawaiian Ridge, although the lack of data makes it difficult to evaluate this in El Hierro) involving a coherent mass of material, which typically deforms during slumping to produce a series of transverse ridges, scarps and folds on the surface of the slump (Moore et al., 1989; Masson et al., 2002). With the occurrence of this type of landslide, the nature of the material involved should give rise to crustal structure with low density. However, the gravity inversion reveals an important low-density zone in the deepest section of the model, on the SE flank of El Hierro, which could be related to a thick layer of clastic sediments derived from the African margin (Schmincke et al., 1995).

The deep, large high-density body at the west and northwest of the island must be cautiously interpreted, because it is located on the edge of the model and not supported by much data. However, this structure corresponds to a stable flank area (Masson et al., 2002) and seems to agree with the interpretation given by Gee et al. (2001) of the NW submarine ridge. This ridge has a highly irregular morphology, characterised by numerous pinnacles, irregular ridges and gullies, that extend over broad area between El Julán and El Golfo landslides, and it is interpreted as a constructional, unfailed island flank. The broad area described on the NW flank contrasts strongly with the narrow, discrete Southern Ridge and its morphology suggests either a wide area of constructional dyke activity or a bifurcation of the onshore rift zones seaward (Gee et al., 2001) (Fig. 5). In the density contrasts model the body associated with this submarine ridge also occupies a wide area and its geometry coincides with the inferred bifurcation pointed out above. The high density in this area is supported by the dyke activity and later stability.

6. Conclusions

The application of a new gravity inversion method described in Montesinos et al. (2005) to the gravity data observed in El Hierro provides a crustal anomalous mass model for this island. The gravity values considered here have been analysed, resulting in a data accuracy level of 0.577 mGal (non-correlated noise). The range of the obtained Bouguer anomaly is 80 mGal, and is used to apply the inversion method.

The adjusted model is constructed by limiting the possible density contrasts between -300 and 300 kg m^{-3}, resulting in a geometrical configuration of high- and low-density areas. The gravity data calculated for this model adjust the observed data with differences according to a Normal distribution, with a standard deviation of 0.805 mGal. The results of the inversion process show a main intrusive body as a result of the accumulation of mantle-type material, which may be related to the earlier shield stage of the island formation. The adjusted low-density bodies appear superimposed on the pre-existing structure and form structures that follow several alignments which we

associate with the recent volcanic activity along the triple-rift system and with the single historic eruption in Lomo Negro.

The gravity inversion model also identifies significant differences between the landslides of El Golfo and of El Julán and Las Playas, with the last one overlying a thick layer of sediments.

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